Spectral–Spatial–Temporal Transformers for Hyperspectral Image Change Detection

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***Abstract*— Convolutional neural networks (CNNs) with excellent spatial feature extraction abilities have become popular in remote sensing (RS) image change detection (CD). However, CNNs often focus on the extraction of spatial information but ignore important spectral and temporal sequences for hyperspectral images (HSIs). In this article, we propose a joint spectral, spatial, and temporal transformer for hyperspectral image change detection (HSI-CD), named SST-Former. First, the SST-Former position-encodes each pixel on the cube to remember the spectral and spatial sequences. Second, a spectral transformer encoder structure is used to extract spectral sequence information. Then, a class token for storing the class information of a single temporal HSI concatenates the output of the spectral transformer encoder. The spatial transformer encoder is used to extract spatial texture information in the next step. Finally, the features of different temporal HSIs are sent as the input of temporal transformer, which is used to extract useful CD features between the current HSI pairs and obtain the binary CD result through multilayer perceptron (MLP). We evaluate the SST-Former on three HSI-CD datasets by numerous experiments, showing that it performs better than other excellent methods both visually and qualitatively. The codes of this work will be available at https://github.com/yanhengwang-heu/IEEE_TGRS_ SSTFormer.**

***Index Terms*— Change detection (CD), cross-attention, hyperspectral image (HSI), self-attention (SA), transformer.**

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I. Introduction

CHANGE detection (CD) by analyzing differences between multitemporal remote sensing (RS) images is the most common and easiest way to monitor large-scale land changes. Hyperspectral images (HSIs) are simultaneously imaged in tens to hundreds of continuous and subdivided spectral bands in the ultraviolet, visible, near-infrared, and mid-infrared regions of the electromagnetic spectrum. The spectral information of HSIs can fully reflect the differences in physical structure and chemical composition within the sample. Therefore, hyperspectral image change detection (HSI-CD) can uniquely provide more subtle and varied changes. In the last few years, HSI-CD has been widely used in military and civilian fields, e.g., agricultural development and damage assessment, which has shown great value, and many state-of-the-art algorithms have been proposed.

The basic flow of CD is as follows. First, RS images of different time phases in the same area are obtained. Second, image preprocessing which can reduce interference caused by different imaging conditions, is used, including geometric correction, registration, and radiometric calibration. Then, the CD algorithm that obtains the CD output is designed. Finally, the output is evaluated. Our study focuses on the CD algorithm, and rests draw on existing research. We categorized HSI-CD algorithms in terms of these topics as follows: 1) algebra-based HSI-CD algorithms; 2) traditional machine learning (TML)-based HSI-CD algorithms; and 3) deep learning (DL)-based HSI-CD algorithms.

Image differencing, image ratioing, and image regression are among the earliest algebra-based CD algorithms. When these methods are applied to HSI-CD, the problem of selecting the optimal band is often faced. Change vector analysis (CVA) which uses all bands to detect change is currently the most commonly applied algebra-based HSI-CD method [1]. Bovolo *et al.* [2] proposed a compressed CVA (C^2VA) which extends CVA to a 2-D representation of change information. Liu *et al.* [3] proposed sequential spectral change vector analysis (S^2CVA) and multiscale morphological compressed change vector analysis (M^2C^2VA) for multiclass CD. There are several other algebra-based CD algorithms, such as Euclidean distance (ED) [4], absolute average difference (AAD) [5], vector angle (VA), and normalized ED (NED) [5]. Algebra-based algorithms often require thresholds to distinguish changing and unchanging pixels. Manual threshold is the earliest and easiest approach, but it is difficult to find appropriate thresholds. Some ML threshold methods that automatically classify change and nonchange are commonly used in conjunction with algebras-based methods, such as expectation maximization (EM) [6], Otsu's method [7], support vector machine (SVM) [8], and K-means [9]. Algebra-based CD algorithms process the pixels of HSIs directly, so they are functional for some spatial textureless HSIs and poor for HSIs with complex spatial texture.

TML-based CD algorithms convert HSIs into feature space to avoid the interference of noisy pixels and contrast features for HSI-CD. Ortiz-Rivera *et al.* [10] used principal component analysis (PCA) to exploit change information in principal components associated with variation. Multivariate alteration detection (MAD) investigates bitemporal relationships by canonical correlation analysis. The CD result of MAD is postprocessed by the maximum autocorrelation factor [11]. Nielsen [12] proposed iteratively reweighted MAD (IR-MAD) for further improving CD accuracy. Wu *et al.* [13] proposed a slow feature analysis (SFA) that trains weights for shrinking the distance of unchanged pixels and expanding the distance of changed pixels. A multilayer cascade screening strategy (MCS^{4}CD) was proposed in [14]. MCS^{4}CD selects the highly reliable unchanged pixels as training samples and extracts the change information by iterative SFA. Some TML-based CD algorithms classify HSIs into objects as detection units. For example, Liu *et al.* [15] counted the subpixel categories within a patch and used the most common category as the patch class for HSI-CD. Nemmour and Chibani [16] used SVM for bitemporal HSI classification and compared classification results for CD. Ahlqvist [17] used the method based on fuzzy sets for land cover classification and took the obtained categories for semantic comparison to obtain CD results. Hussain *et al.* [18] summarized object-based CD algorithms in 2013. Currently, TML is one of the most popular approaches for HSI-CD.

DL is a stronger feature extraction algorithm than TML in most vision tasks. DL has been widely used in HSI processing, e.g., classification [19], [20], [21], [22], [23], [24], [25], [26], [27], unmixing [28], [29], and anomaly detection [30], [31]. At the same time, there have been many well-recognized works on DL-based HSI-CD, such as convolutional neural networks (CNNs), 3-D CNNs, autoencoders, recurrent neural networks (RNNs), graph convolutional networks (GCNs) [32], generative adversarial networks (GANs), and attention networks, and they have achieved a good performance. Li *et al.* [33] designed an unsupervised CNN trained by highly reliable results of CVA and structural similarity (SSIM). Wang *et al.* [34] proposed a general end-to-end 2-D CNN (GETNET) that combines unmixing and CNN. Yang *et al.* [35] used a residual attention and a 3-D multiscale pyramid CNN for HSI-CD. Lopez-Fandino *et al.* [36] used the watershed transform to obtain a binary CD map, extracted the features of bitemporal HSIs by stacked autoencoders, and classified the change pixels of a binary CD map for multiclass CD by SVM and extreme learning machine (ELM). Zheng *et al.* [37] innovatively proposed a cross-resolution difference learning method to achieve amazing performance from different resolution images, without requiring resizing operations. Zhao *et al.* [38] designed a 3-D autoencoder to obtain spectral–spatial features and classified them by a 2-D CNN. Song *et al.* [39] used PCA and the spectral correlation angle (SCA) to generate training pixels with high probabilities of being changed or unchanged for unsupervised CD and proposed a recurrent 3-D fully convolutional network (Re3FCN). The Re3FCN includes a 3-D CNN for extracting spatial and spectral information and long short-term memory (LSTM) for extracting change information. A dual-branch difference amplification GCN (D^{2}AGCN) was proposed in [40]. D^{2}AGCN extracts non-Euclidean information of bitemporal HSIs via GCN and makes a difference amplification module to emphasize the change features. An unsupervised CD-GAN that predesigned and trained a deep adversarial network was designed in [41]. Wang *et al.* [42] presented a spectral–spatial-wise attention (SSA-SiamNet) for HSI-CD. The spectral attention module and spatial attention module are used to extract features of multitemporal HSIs and obtain the CD map by ED. Although the above DL-based algorithms have achieved good CD results, they still have the following limitations.

1) HSIs are spectral sequence signals from which one continuous electromagnetic spectrum is measured. The spectral sequence information is worth analyzing. However, CNNs treat a spectrum as a processing unit and do not analyze information from each band and the sequence of bands in the spectrum. CNNs may learn the above information through their powerful feature extraction ability, but it is not interpretable. RNNs can model the spectral sequence but are prone to gradient loss for long sequences [43].
2) The location information of objects in space is very important. CNNs extract features through a sliding convolution kernel and do not consider the position of the convolution kernel as it slides over the image. In addition, the spatial position of pixels in a convolution kernel is not considered.
3) The temporal sequence information can be regarded as the mapping relationship between the same objects or different objects in bitemporal HSIs. The above algorithm uses difference or concatenation to compare the characteristics of bitemporal HSIs and does not learn the temporal sequence information. This can be achieved using RNNs, but their image feature extraction is poor.

Transformers that can model long sequence dependencies well have become popular in computer vision tasks, such as image classification [44], [45], object detection [46], and text detection [47]. They have also been successfully applied in RS image processing, for example, HSI classification [43], [48], high-resolution RS CD [49], and HSI super-resolution [50]. Transformers rely entirely on self-attention (SA) to model global dependencies on the input. Position embedding can be used to determine the absolute position of each pixel. Transformers can learn complex RS image spatial relationships well, establishing connections between nonadjacent objects. However, the above transformer methods only encode spatial sequences or spectral sequences. In this article, we propose a joint spectral, spatial, and temporal transformer (SST-Former) for HSI-CD. SST-Former has a spectral–spatial transformer

(SS-Former) with a dual-branch structure, which can extract bitemporal HSI features. A temporal transformer (T-Former) is designed to learn the temporal features and obtain the CD results. The main contributions of the proposed SST-Former can be summarized as follows.

1) SST-Former, an end-to-end transformer model, is proposed for HSI-CD. It simultaneously considers the spatial, spectral, and temporal information of HSIs. SST-Former can solve the above three limitations of existing DL-based algorithms. Verification on three widely used hyperspectral datasets shows that the transformer is suitable for HSI-CD with limited training samples for the first time.
2) SS-Former, which uses three transformer encoders of spectral bands to extract the sequence information of spectra and uses two transformer encoders of spatial pixels to extract the sequence information of space, is designed. SS-Former spectrally and spatially learns the long-distance dependence from HSIs, takes into account the spectral–spatial sequence and location information, and can more fully extract the feature information of HSIs.
3) T-Former, which uses three residual structures with cross attention to integrate the class token of one temporal HSI and the patch token at another temporal HSI to obtain the temporal sequence information, is designed.

II. Methodology

In this section, we first introduce some preliminaries of transformers. Then, the proposed SST-Former with two well-designed modules, SS-Former and T-Former, is presented. The flowchart of the proposed SST-Former is shown in Fig. 1. Detailed explanations of these parts are given in Sections II-A–II-C, respectively.

A. Review of Transformers

Transformers have gradually replaced RNNs in the fields of natural language processing (NLP) and machine translation. Transformers are also explored in computer vision. There are many transformer structures for different tasks, e.g., multimodal video transformer (MM-ViT) for video action recognition [51], ViT-YOLO for object detection [52], and spectral transformer for HSI classification [43]. What is a transformer? In general, transformers are the structures that can model long-sequence information through the three main modules of position embedding, SA, and multilayer perceptron (MLP).

Position embedding captures the order of an input sequence. In [53], position embedding was generated by sine and cosine functions of different frequencies. Gehring *et al.* [54] used trainable weights as position embedding. They embedded the input elements $\boldsymbol{x} = (x_1, \ldots, x_n)$ into the distribution space $\boldsymbol{w} = (w_1, \ldots, w_n)$, where n is the number of input elements. Then, they also equipped the model with positional embedding $\boldsymbol{p} = (p_1, \ldots, p_n)$, and the two were combined to obtain the input element representation $\boldsymbol{e} = (w_1 + p_1, \ldots, w_n + p_n)$. $\boldsymbol{e}$ and $\boldsymbol{p}$ were finally determined by model training.

SA builds a relationship of all input elements. Fig. 2(a) shows the workflow of SA with n input elements and output only for the first element. Each element has three features, i.e., query, key, and value, for remembering the relationships between them. Query is used to match others. Key is used for matching. Value is the information to be extracted. To improve the feature extraction ability, an improved SA, multihead self-attention (MSA), is proposed. The first head is used to extract feature one, the second head is used to extract feature two, and so on. We take two heads as an example in Fig. 2(b) to show MSA. For details, please refer to [53].

An MLP composed of fully connected layers, and a rectified linear unit (ReLU) activation function is used as a feedforward network in the transformer encoder. The MLP also plays a nonlinear role. Another function is classifying in transformer decoder, named MLP head. The MLP head is composed of only a fully connected layer and a layer normalization. Therefore, MLP is indispensable for transformers.

B. Spectral–Spatial Transformer

Existing transformers encode only spatial sequences of multispectral images (MSIs) in image processing, such as vision transformer (ViT) [44]. Approximately, continuous spectral signatures are also important for HSIs. Hence, SS-Former is proposed to extract HSI spectral–spatial features and is depicted in Fig. 1.

First, we divide an HSI pair into many patch pairs. Some patch pairs are used for training SST-Former, and others are used for testing. Suppose a patch pair is $\boldsymbol{X} = \{X_1, X_2\}$. For one branch, we reshape a patch $X \in \mathbb{R}^{w\times h\times c}$ into a spectral sequence set $\boldsymbol{x} = \{x_1, x_2, \ldots, x_c\}$, where w, h is the spatial resolution of the patch and c is the number of channels. For the convenience of calculation, we regard $\boldsymbol{x}$ as a matrix of $c \times wh$. n dimensions constant latent vectors obtained by linear projection are formulated by

$$\boldsymbol{E} = \omega_1 \boldsymbol{x}, \quad \omega_1 \in \mathbb{R}^{wh\times n} \tag{1}$$

where ω_1 is the trainable weight matrix with size $wh \times n$ and $\boldsymbol{E} \in \mathbb{R}^{c\times n}$ collects the linear projection features of spectral bands. Position embedding $\boldsymbol{p}$ is combined with $\boldsymbol{E}$ by

$$\boldsymbol{Z}_0 = \boldsymbol{E} + \boldsymbol{p}, \quad \boldsymbol{E} \in \mathbb{R}^{c\times n}, \ \boldsymbol{p} \in \mathbb{R}^{c\times n} \tag{2}$$

where $\boldsymbol{p} = (p_1, \ldots, p_n)$ is regarded as a $c \times n$ random matrix and is determined by model training. $\boldsymbol{Z}_0$ is fed into the transformer encoder composed of alternating layers of MSA and MLP to encode the information of the spectral sequence, which can be formulated as follows:

$$\boldsymbol{Z}'_l = \mathrm{MSA}(\mathrm{LN}(\boldsymbol{Z}_{l-1})) + \boldsymbol{Z}_{l-1}, \quad l \in \{1, 2\} \tag{3}$$

$$\boldsymbol{Z}_l = \mathrm{MLP}\big(\mathrm{LN}\big(\boldsymbol{Z}'_l\big)\big) + \boldsymbol{Z}'_l, \quad l \in \{1, 2\} \tag{4}$$

where LN is the layernorm. The number of spectral transformer encoders is 2, which means that (3) and (4) are recycled two times. $\boldsymbol{Z}_2$ is the final output of the spectral transformer encoders.

Then, we reshape the output of the spectral transformer encoder $\boldsymbol{Z}_2 \in \mathbb{R}^{c\times n}$ into $\boldsymbol{Z}_2 \in \mathbb{R}^{n\times c}$. m dimensions constant latent vectors obtained by linear projection are formulated by

$$\boldsymbol{Z} = \omega_2 \boldsymbol{Z}_2, \quad \omega_2 \in \mathbb{R}^{c\times m} \tag{5}$$

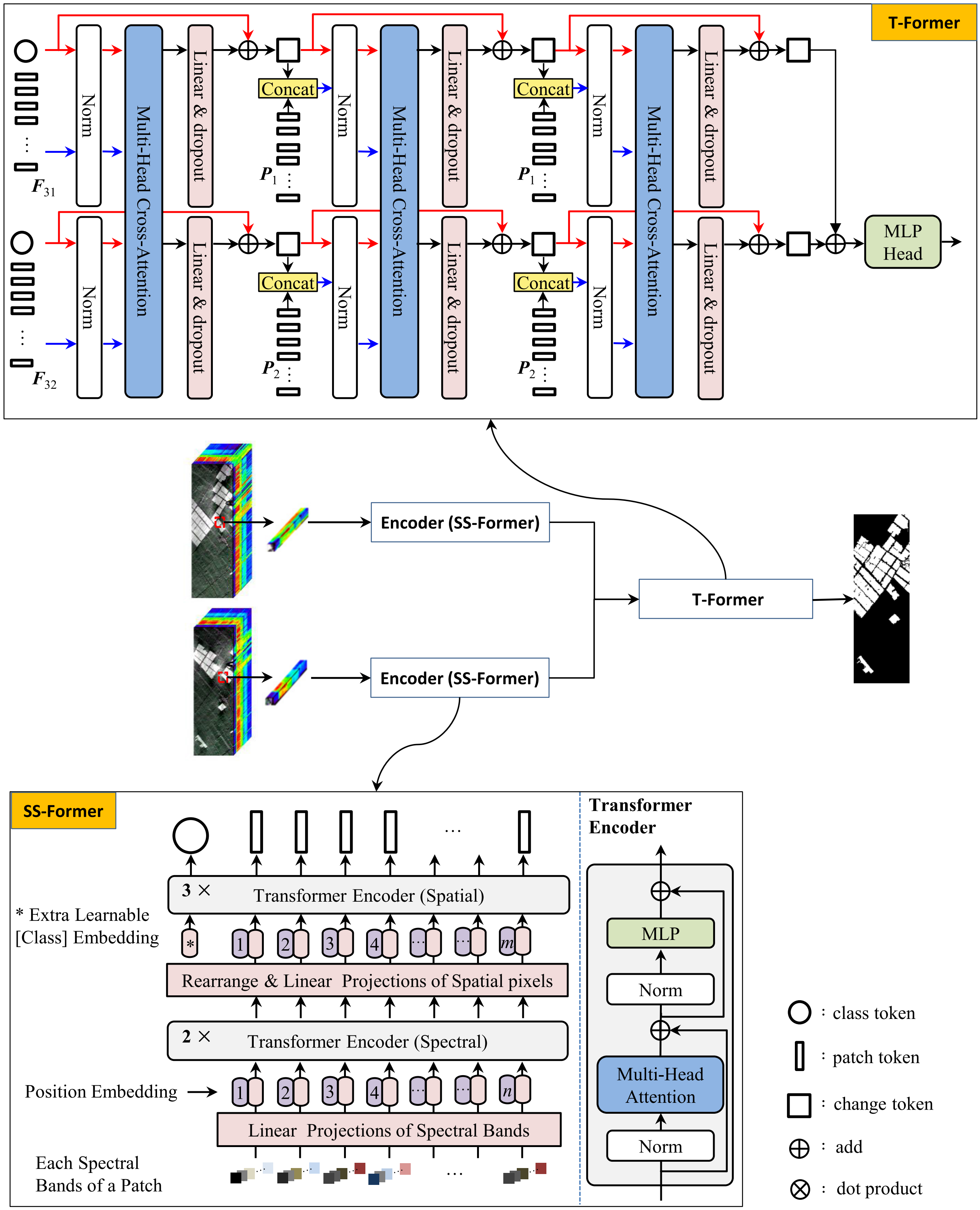
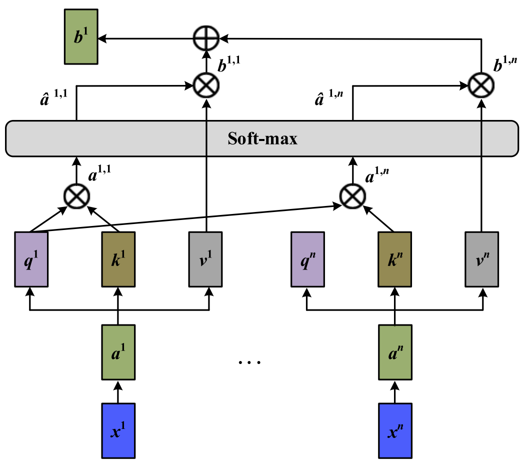
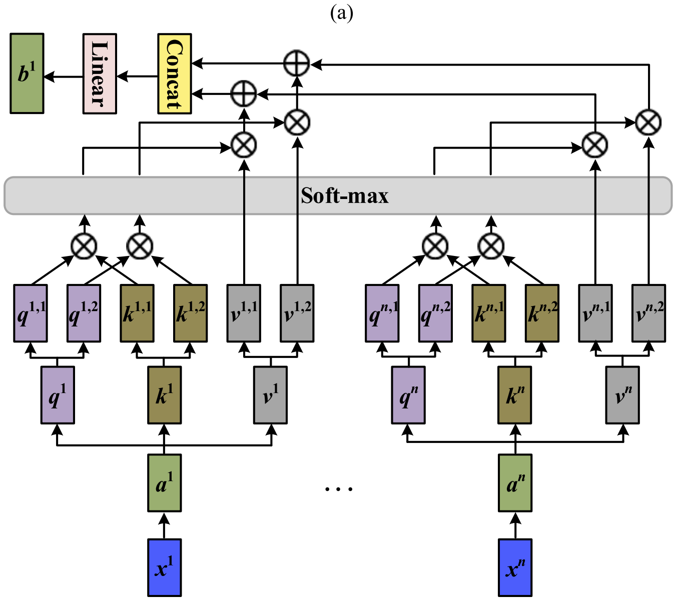


Fig. 1. Network structure of the proposed SST-Former for HSI-CD. SST-Former consists of SS-Former and T-Former. The transformer encoder (spectral) and the transformer encoder (spatial) have the same network process but different parameters in SST-Former. The red arrow means that input is related only to a class token or a change token. The blue arrow means that input is related to patch tokens concatenated with a class token or a change token.

where ω_2 is a trainable weight matrix with size $c \times m$ and $\boldsymbol{Z} \in \mathbb{R}^{n \times m}$ collects the output features of linear projection of spatial pixels. Similar to bidirectional encoder representation from transformer (BERT) [55] and ViT [44], an extra learnable



(a)



(b)

Fig. 2. Illustration of: (a) SA and (b) MSA. This is a simple example of n inputs and one output, and the output of other elements involves the same process. q, k, and v refer to query, key, and value, respectively.

class embedding $C \in \mathbb{R}^{1\times m}$ is prepended. C can learn patch pseudo-category through transformer encoder. First, we concatenate a variable $\boldsymbol{F}_0 \in \mathbb{R}^{(n+1)\times m}$ between C and $\boldsymbol{Z}$. $\boldsymbol{F}_0$ can be formulated by

$$\boldsymbol{F}_0 = \mathrm{Concat}(C, \boldsymbol{Z}). \quad (6)$$

Next, we feed $\boldsymbol{F}_0$ into the spatial transformer encoder which has the same structure as the spectral transformer encoder but different parameters for modeling the spatial sequence feature. It can be summarized by

$$\boldsymbol{F}'_d = \mathrm{MSA}(\mathrm{LN}(\boldsymbol{F}_{d-1})) + \boldsymbol{F}_{d-1}, \quad d \in \{1, 2, 3\} \quad (7)$$

$$\boldsymbol{F}_d = \mathrm{MLP}(\mathrm{LN}(\boldsymbol{F}'_d)) + \boldsymbol{F}'_d, \quad d \in \{1, 2, 3\} \quad (8)$$

where the number of spatial transformer encoders is 3, which means that (7) and (8) are recycled three times. $\boldsymbol{F}_3 \in \mathbb{R}^{(n+1)\times m}$ is the final output of spatial transformer encoders. We take the first sequence of $\boldsymbol{F}_3$ as the class token $C \in \mathbb{R}^{1\times m}$ and the second to nth sequence of $\boldsymbol{F}_3$ as patch tokens $\boldsymbol{P} \in \mathbb{R}^{n\times m}$. The class token can encode the statistical characteristics of an entire HSI patch and avoid bias toward a particular token in the sequence because it is not based on the image content itself. Therefore, it can be seen a pseudo-category of patch. Different from ViT, there is no position embedding for the class token because it is always the first sequence of $\boldsymbol{F}_3$. Patch token is the deep feature combined spectrum and space of the patch.

We share the parameters of dual-branch SS-Formers in the network, except for extra learnable class embedding. There are two extra learnable class embedding and shared a position embedding in dual-branch SS-Formers. Finally, we can obtain two outputs $\boldsymbol{F}_{31} = \{C_1, \boldsymbol{P}_1\}$ and $\boldsymbol{F}_{32} = \{C_2, \boldsymbol{P}_2\}$.

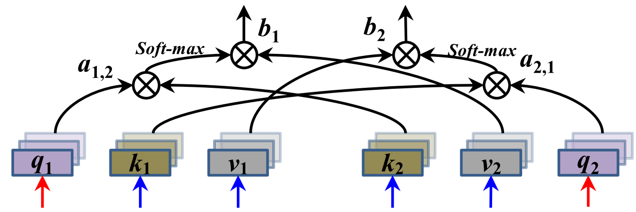


Fig. 3. Illustration of MCA. ⊗ indicates the dot product.

C. Temporal Transformer

Difference and concatenation methods are commonly used to address the relationship between two temporal RS images. They only compare the characteristics between two temporal RS images and do not consider the time sequence about change rule. The transformer has a powerful sequence encoding ability, and it can learn the change-related features and obtain the change rules between the objects of multitemporal HSIs. Hence, we design T-Former to model the time sequence for HSI-CD.

T-Former contains a three-layer multihead cross attention (MCA) and an MLP structure. It can be seen in Fig. 1. The CA is similar to an SA with two inputs. Unlike SA, the query of input one just dot product the key and value of input two. For details of MCA, please refer to Fig. 3.

After SS-Former, we obtain the query only from the class token C and the key and the value from $\boldsymbol{F}_3$. The calculation formulas of query, key, and value are as follows:

$$q = W_q \times C, \quad q \in \mathbb{R}^{1\times m}, \quad W_q \in \mathbb{R}^{m\times m} \quad (9)$$

$$k = W_k \times \boldsymbol{F}_3, \quad k \in \mathbb{R}^{(n+1)\times m}, \quad W_k \in \mathbb{R}^{m\times m} \quad (10)$$

$$v = W_v \times \boldsymbol{F}_3, \quad v \in \mathbb{R}^{(n+1)\times m}, \quad W_v \in \mathbb{R}^{m\times m} \quad (11)$$

where W_q, W_k, and W_v are the weights for getting query, key, and value. r and m/r are the head number and head dimension of q, k, and v. For bitemporal HSIs, we can obtain q_1, q_2, k_1, k_2, and v_1, v_2 with share weights. Then, q_1 dot product k_2 and multiplied by scale s to get hidden feature $a_{1,2}$, where s is $\sqrt{m/r}$. After soft-max, $a_{1,2}$ dot product v_2 to get feature b_1. Similarly, we can obtain b_2, $b_1, b_2 \in \mathbb{R}^{1\times m}$.

For T-Former, we layernorm $\boldsymbol{F}_{31}$ and $\boldsymbol{F}_{32}$ and feed them into the first MCA to obtain b_{11} and b_{12}. b_{ij} is the output obtained after the ith MCA in branch j. Linear projection with a dropout layer is used to optimize the feature. To prevent information loss and gradient decay, a residual structure is designed. The first change token $T_{1j} \in \mathbb{R}^{1\times m}$ is obtained as follows:

$$b_{1j} = \mathrm{MCA}(\boldsymbol{F}_{3j}) \tag{12}$$

$$T_{1j} = \mathrm{Dt}(\omega_3 b_{1j}) + C_j, \quad \omega_3 \in \mathbb{R}^{m\times m} \tag{13}$$

where Dt (x) is the dropout layer applied to prevent overfitting and $j \in \{1, 2\}$ is the branch of bitemporal HSIs. ω_3 is the parameter of linear projection. Then, we obtain the query of second MCA from T_{1j}, and the key and value from T_{1j} concatenated with $\boldsymbol{P}_j$. The above process is repeated two times, as follows:

$$b_{ij} = \mathrm{MCA}(\mathrm{Concat}(T_{(i-1)j}, \boldsymbol{P}_j)), \quad i \in \{2, 3\} \tag{14}$$

$$T_{ij} = \mathrm{Dt}(\omega_{i+2} b_{ij}) + T_{(i-1)j}, \quad \omega_{i+2} \in \mathbb{R}^{m\times m} \tag{15}$$

where Concat is the abbreviation for concatenation. After the 2th (13) and (14), we can obtain the final change tokens T_{31} and T_{32}. They are added to T_3, and the CD result is obtained by the MLP head with layernorm and one linear layer

$$y' = w\mathrm{LN}(T_3), \quad w \in \mathbb{R}^{m\times 2} \tag{16}$$

where $y' \in \mathbb{R}^{1\times 2}$ is the output of the MLP head and w is the weight of the linear layer. If y' (0,0) $>$ y' (0,1), the CD result of the patch pair is unchanged. Otherwise, it is changed.

III. Experiments

In this section, we evaluate and analyze the influence of SST-Former via experiments. First, we provide a detailed description of the experimental datasets. Second, we describe the experimental settings, including the evaluation criteria, comparative experiment, hyperparameter of SST-Former, and laboratory environment. Third, we compare performance between the proposed method and other state-of-the-art CD algorithms. Fourth, we perform ablation analysis to reveal the innovations of SS-Former and T-Former. Fifth, parameters of SST-Former are analyzed.

A. Experimental CD Datasets

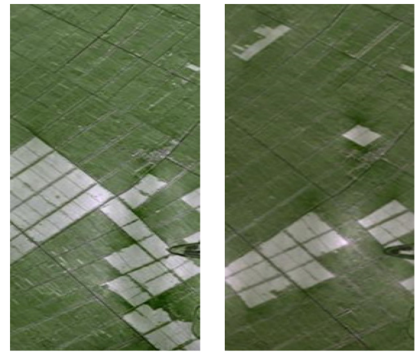
Three CD datasets are used to evaluate the performance of SST-Former, including the Farmland CD dataset,[1] Barbara CD dataset,[2] and Bay Area CD dataset.[3] The details of the three datasets are as follows.

1) Farmland Dataset: The Farmland dataset is also known as the China dataset. The bitemporal HSIs were taken on May 3, 2006 and April 23, 2007. The cover land is farmland in Yancheng, Jiangsu, China. The satellite sensor is the Earth Observing-1 (EO-1) Hyperion hyperspectral sensor. The size of the bitemporal HSIs is 420 × 140 pixels with a 30-m spatial resolution. The dataset contains 155 bands after removing the noise and water absorption bands in the spectral range of 0.4–2.5 μm. It mainly includes changes in corps and can be used in cultivated land change surveys. The pseudo-color images with bands 91, 103, and 123 are shown in Fig. 4.

[1]https://rslab.ut.ac.ir/data

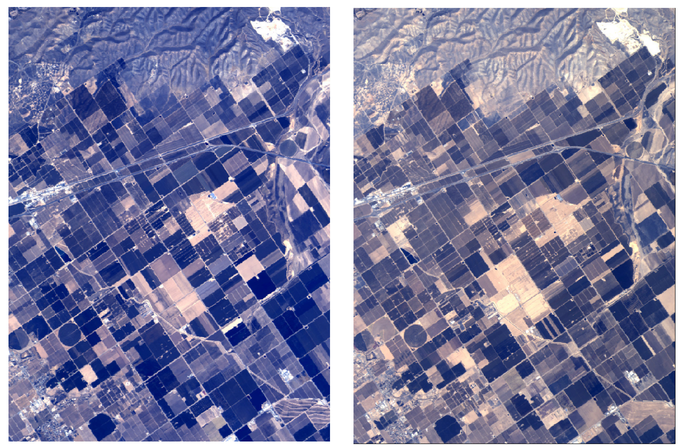
[2]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

[3]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset



(a) (b)

Fig. 4. Farmland dataset: (a) image acquired on May 3, 2006 and (b) image acquired on April 23, 2007.



(a) (b)

Fig. 5. Santa Barbara dataset: (a) image acquired in 2013 and (b) image acquired in 2014.

2) Santa Barbara Dataset: The bitemporal HSIs contained in the Santa Barbara dataset were obtained in 2013 and 2014, as shown in Fig. 5, using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The covered land is Santa Barbara, CA, USA. The two HSIs have the same size of 984 × 740 pixels with a spatial resolution of 20 m. After noisy band removal, they are characterized by 224 bands covering the spectral range of 0.4–2.5 μm. In this dataset, there are 52 134 changed pixels, 80 418 unchanged pixels, and 595 608 unknown pixels.

3) Bay Area Dataset: The Bay Area dataset with bitemporal HSIs was obtained in 2013 and 2014 by the AVIRIS sensor. It covers an area of Patterson in CA. The size of the dataset is 600 × 500 pixels with a 20-m spatial resolution. It covers 224 spectral bands in the range of 0.4–2.5 μm. In this dataset, there are 38 425 changed pixels, 34 211 unchanged pixels, and 227 364 unknown pixels. RGB images from the Bay Area dataset are shown in Fig. 6.

To obtain better result with a limited sample, we selected 500 changed pixels and 500 unchanged pixels to train SST-Former on three datasets. The numbers of training pixels and test pixels are listed in Table I.

(a) (b)

Fig. 6. Bay Area dataset: (a) image acquired in 2013 and (b) image acquired in 2014.

B. Experimental Setup

Overall accuracy (OA) and kappa coefficient (Kappa) were utilized as quantitative indicators, which are calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{17}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{Pc}}{1 - \mathrm{Pc}} \tag{18}$$

$$\mathrm{Pc} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2} \tag{19}$$

where true positive (TP) is the number of pixels that change in ground truth and also change in the predicted image, true negative (TN) indicates the number of pixels that do not change in ground truth and do not change in the predicted image, false positive (FP) represents the number of pixels that do not change in ground truth but do change in the predicted image, and false negative (FN) is the number of pixels that change in ground truth but do not change in the predicted image.

Six excellent algorithms are selected for comparison with and to verify the effectiveness of the proposed algorithm, including CVA, differential principal component analysis (DPCA) [56], SVM, light CNN (LCNN), GETNET [34], Re3FCN [39], and ViT [44]. The patch size is 5 expect for CVA and DPCA. The other details of these algorithms are as follows.

CVA was used to obtain the change vector and automatically classified change and nonchange by K-means.

DPCA was used to get the difference map of bitemporal HSIs and detected changes by K-means. MATLAB CD Toolbox[4] was selected for the implementation of CVA and DPCA.

For the SVM, PCA was used to extract the ten principal components of the difference map, and 500 changed pixels and 500 unchanged pixels were used to train the SVM. This was executed by sklearn.svm in Python.

The LCNN itself consisted of three convolutional layers. The kernel size of the first two convolution layers was 3×3, followed by a ReLU activation function, and the kernel size of the third layer was 1×1. The channel numbers of the three convolutional layers were 256, 128, and 2. The step size was 1, and the padding was 0.

For GETNET, Wang *et al.* [34] used a large amount of data to train the model. First, HSIs were processing with linear mixing methods [28], [57]. Then, we needed only 500 changed pixels and 500 unchanged pixels to train GETNET and obtain a good result. We set the epoch to 1000. The other parameters were the same as those in the original article.

For Re3FCN, there are two convolutional layers for extracting spatial features and two recurrent layers with LSTM for time sequence features. The details can be found in this article and code.[5] The numbers of changed pixels and unchanged pixels were also 500. For the Farmland and Santa Barbara datasets, we layer normalized the input. The Bay Area dataset was not normalized.

For the ViT, we used a two-branched structure, and class tokens of the bitemporal HSIs were subtracted and concatenated for CD. ViT performs transformer encoding only on spatial information. On this basis, we propose the SST-Former algorithm in this article. At the same time, we converted the ViT spatial transformer encoding into spectral transformer encoding and combined difference algorithm and T-Former to verify the effect of SST-Former.

We implemented our proposed method on PyTorch 1.10.1 with an Intel Core i9-10900k CPU and an NVIDIA GeForce RTX 3060 12 G GPU. The Adam optimizer was used with a learning rate of 0.0005. The learning rate was simultaneously decayed by multiplying 0.9 after each 1/20th of the total epoch. The total number of epochs was 400. ReLU was used as the activation function, and the loss function was cross-entropy. The 5 and 64 were the sizes of the patch and batch, respectively. The dimension n of spectral linear projection was 32. The dimension m of spatial linear projection was 256. The head numbers of spectral MSA, spatial MSA, and temporal MCA were 4, 8, and 8, respectively. The dropout layer was employed to inhibit 20% of neurons.

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THREE DATASETS

	Unchanged		Changed	
	Training	Test	Training	Test
Farmland	500	39917	500	17 883
Santa Barbara	500	79918	500	51634
Bay Area	500	33711	500	37925

C. Comparison With Other Well-Established Methods

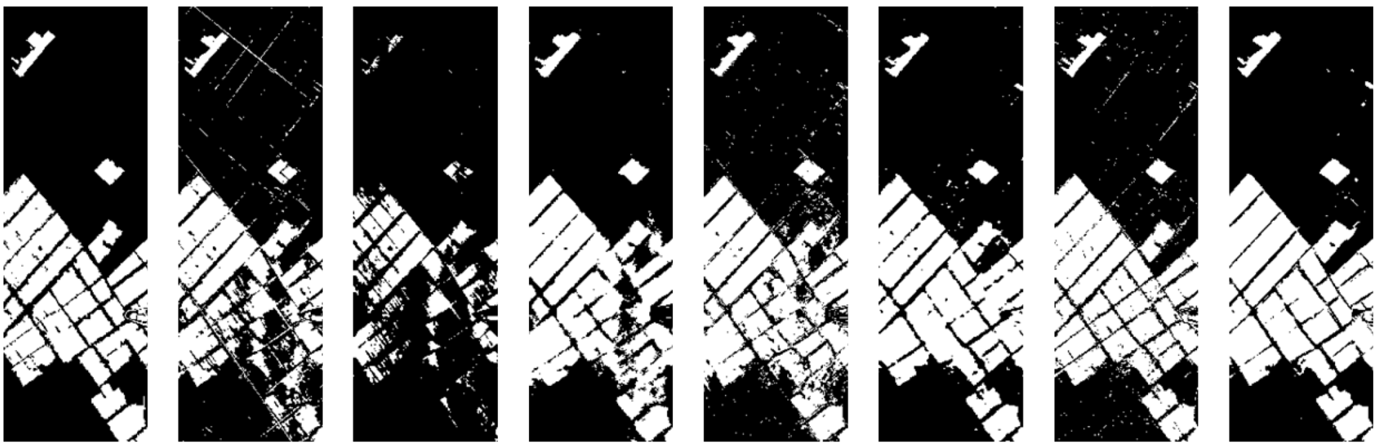
To evaluate the effect of SST-Former, we compared our method with other well-established methods both visually and qualitatively. All methods were tested five times with a different training set. Table II gives the detailed OA and

[4]https://github.com/Bobholamovic/ChangeDetectionToolbox

[5]https://github.com/mkbensalah/Change-Detection-in-Hyperspectral-Images

TABLE II
ACCURACY COMPARISON OF EXCELLENT HSI-CD METHODS ON THREE DATASETS

		CVA	DPCA	SVM	LCNN	Re3FCN	GETNET	Proposed
Farmland	Kappa	0.6998	0.4809	0.7897	0.7761	0.8828	0.864	**0.8843**
	OA	0.8749	0.8100	0.9087	0.9012	0.9490	0.9403	**0.9521**
Santa Barbara	Kappa	0.6519	0.4403	0.9377	0.9156	0.9509	0.9448	**0.9658**
	OA	0.8325	0.7566	0.9701	0.9598	0.9766	0.9737	**0.9810**
Bay Area	Kappa	0.7063	0.3790	0.8094	0.9328	0.8844	0.9472	**0.9594**
	OA	0.8517	0.6777	0.9051	0.9665	0.9424	0.9749	**0.9798**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. CD results on the Farmland dataset: (a) ground truth; (b) CVA; (c) DPCA; (d) SVM; (e) LCNN; (f) Re3FCN; (g) GETNET; and (h) SST-Former. The white area is changed, and the black area is unchanged.

Kappa information for the three datasets. Figs. 7–9 show the visualization results for the Farmland, Santa Barbara, and Bay Area datasets, respectively.

1) Results Analysis for the Farmland Dataset: The unsupervised algebra-based CVA algorithm can achieve 0.8749 and 0.6998 on OA and Kappa, respectively. The lowest OA and Kappa are achieved using the unsupervised TML-based method DPCA. In relation to vision, CVA and DPCA had many missing alarm areas, and CVA also had false alarm areas. Supervised learning-based SVM greatly improved accuracy and presented a good visual for HSI-CD. However, SVM generated clear areas of missing alarms and false alarms, too. DL methods, e.g., GETNET and Re3FCN, were 10% about higher than SVM in terms of Kappa, except that LCNN. GETNET and Re3FCN showed only a small amount of salt and pepper noise in vision. SST-Former achieved higher CD accuracy than other DL-based algorithms. In Fig. 7(h), SST-Former yielded less salt and pepper noise than GETNET and Re3FCN. However, some unchanged areas mixed in with the changed areas were misclassified.

2) Results Analysis for the Santa Barbara Dataset: Extremely low Kappa was obtained by DPCA. CVA had a good performance than DPCA. In Fig. 8(b) and (c), the short CVA CD map was dominated by false alarm areas, and that of DPCA was dominated by missing alarm areas. SVM had a better accuracy than CVA and DPCA. SVM had false alarm areas on object edges in Fig. 8(d). Except for that of LCNN, the accuracy of DL-based methods, e.g., Re3FCN and GETNET, was approximately 1% higher than that of SVM in terms of OA and Kappa. They also had better visual effects. Compared to other algorithms, SST-Former achieved higher accuracy. It also yielded better visual results with sporadic salt and pepper noise, as shown in Fig. 8(h).

3) Results Analysis for the Bay Area Dataset: CVA and DPCA, similarly, had low accuracy (as shown in Table II) and a high missing rate and a high false rate (as shown in Fig. 9). SVM showed a qualitative improvement in quality and vision. LCNN performed approximately 13% better than SVM in terms of Kappa, and GENNET performed better than LCNN while Re3FCN performed worse than LCNN, in contrast to the results for the Farmland and Santa Barbara datasets. The DL-based methods all showed sporadic salt and pepper noise in Fig. 9 but as much as the SVM. Our proposed SST-Former was observed to achieve a higher CD accuracy and more desirable CD map than other state-of-the-art methods.

In conclusion, DPCA performs the worst on the three datasets. CVA is worth studying in unsupervised HSI-CD tasks, but the performance gap is still large compared to that of supervised algorithms. SVM showed a dramatic improvement but was still inferior to the DL-based methods. DL-based methods extract the deep features of HSIs through powerful nonlinear feature extraction, which further improves the CD effect. Using a transformer to extract spectral, spatial, and temporal sequence information, SST-Former had higher accuracy than the other methods, and the generate map had fewer noisy points. Therefore, our proposed SST-Former extracts

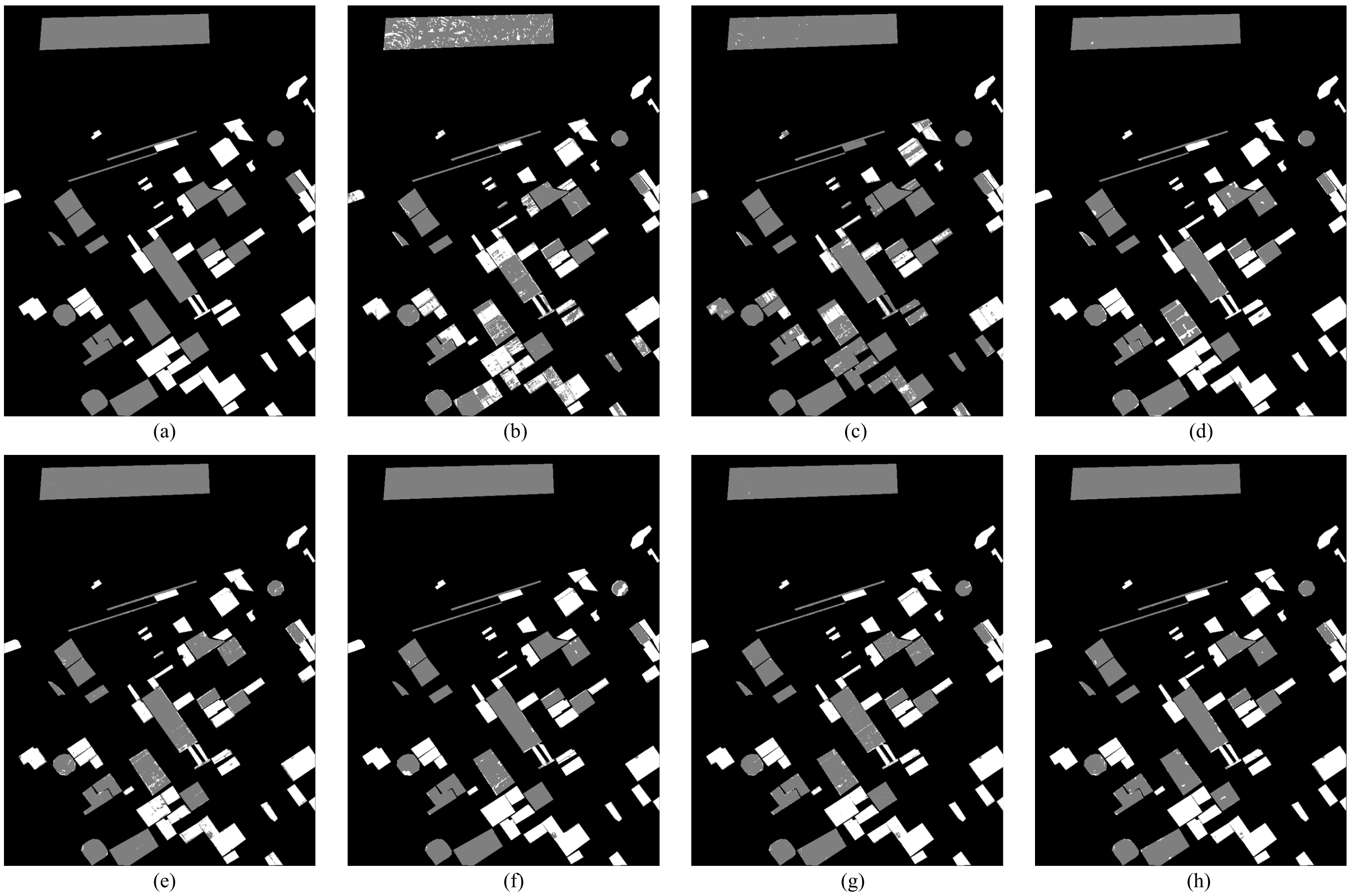


Fig. 8. CD results of on the Santa Barbara dataset: (a) ground truth; (b) CVA; (c) DPCA; (d) SVM; (e) LCNN; (f) Re3FCN; (g) GETNET; and (h) SST-Former. The white is the changed area. The gray is the unchanged area. The black is the unknown area.

spectral, spatial, and temporal information and is more efficient for HSI-CD.

D. Ablation Analysis

Our proposed SST-Former has two innovation points: SS-Former and T-Former. Ablation analysis was carried out to verify the effectiveness of these two points.

For SS-Former, we set ViT as a dual-branch structure and made two class tokens of bitemporal HSIs subtract for CD. ViT encodes only the spatial sequence information of HSIs, ViT-spatial for short. We converted the spatial encoding of ViT to spectral encoding, named ViT-spectral. A comparison of ViT-spatial + difference, ViT-spectral + difference, and SS-Former + difference was performed, and the evaluation results for Kappa are shown in Fig. 10(a). Except on the Santa Barbara, SS-Former + difference performed better than the other methods with difference. At the same time, three different encoders with concatenation were performed. In Fig. 10(b), we find that ViT-spatial + concatenation is better than SS-Former + concatenation about 0.002 on Farmland, and SS-Former + concatenation is better on the other datasets. We compared ViT-spatial + T-Former, ViT-spectral + T-Former, and SS-Former + T-Former (SST-Former), and the results are shown in Fig. 10(c). The Kappa of SST-Former was superior to those of other approaches with T-Former on the three datasets. Therefore, SS-Former is better than ViT-spectral and ViT-spatial, and can more effectively extract spatial and spectral sequence features and improve the accuracy of CD.

For T-Former, we compared the results of using the difference operator, concatenation operator, or T-Former on the ViT-spectral, ViT-spatial, and SS-Former encoder structures. Fig. 11 shows the Kappa of the comparison results on the three datasets. In Fig. 11(a), the difference effect is the worst, the concatenation effect is the best, and T-Former has a similar performance to concatenation. In Fig. 11(b), the effect of the difference is better than that of the concatenation and is not as good as T-Former. In Fig. 11(c), the effect of difference and concatenation is similar but not as good as that of T-Former. Therefore, the T-Former is a more stable and effective feature comparison algorithm on different datasets and can learn the rule of change in the temporal sequence for HSI-CD.

E. Effect of Parameters

The performance of SST-Former relies on the quality of the parameters. Therefore, the influence of the parameters training set number, patch size, dimension n, dimension m, number of spectral transformer encoders, number of spatial transformer encoders, and number of temporal CA layers are investigated. The Kappa results on the three datasets are shown in Fig. 12.

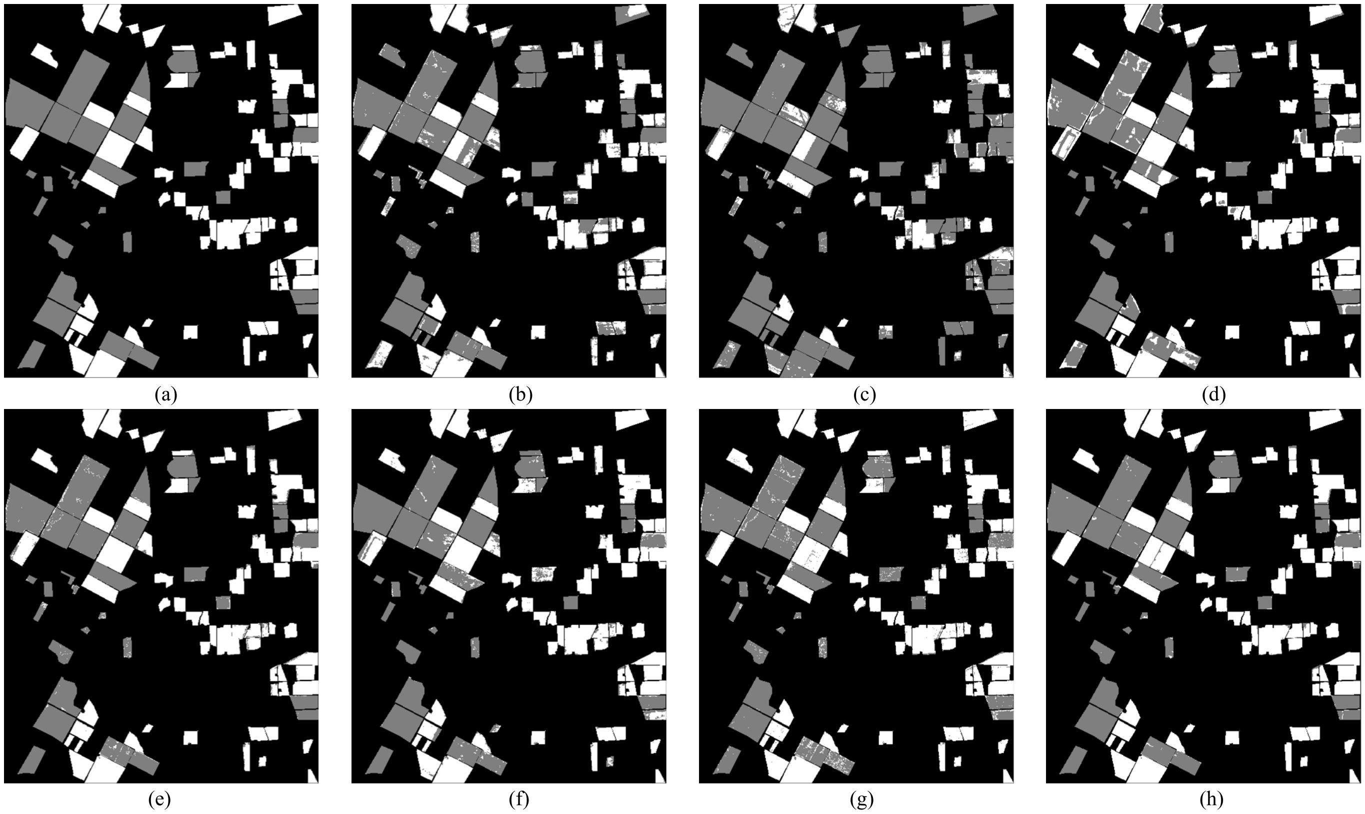


Fig. 9. CD results on the Bay Area dataset: (a) ground truth; (b) CVA; (c) DPCA; (d) SVM; (e) LCNN; (f) Re3FCN; (g) GETNET; and (h) SST-Former. The white is the changed area. The gray is the unchanged area. The black is the unknown area.

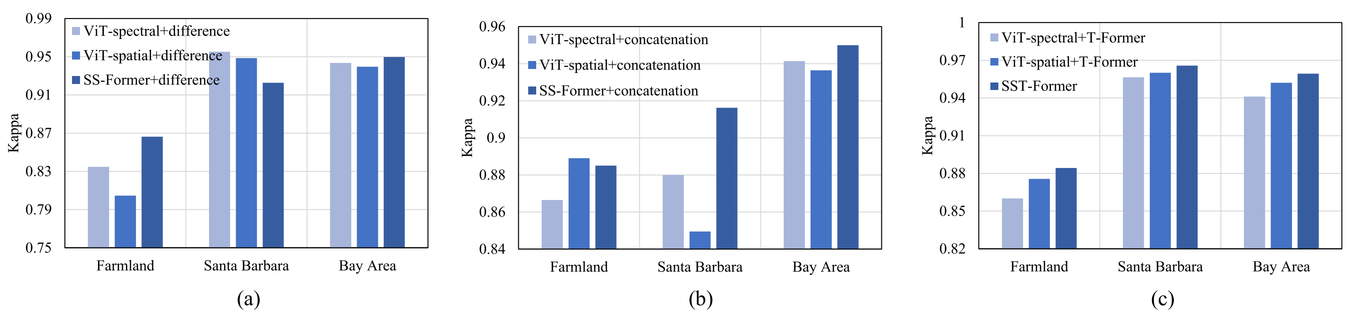


Fig. 10. Evaluation of the Kappa for SS-Former on three datasets: (a) ViT-spectral, ViT-spatial, and SS-Former with difference; (b) ViT-spectral, ViT-spatial, and SS-Former with concatenation; and (c) ViT-spectral, ViT-spatial, and SS-Former with T-Former.

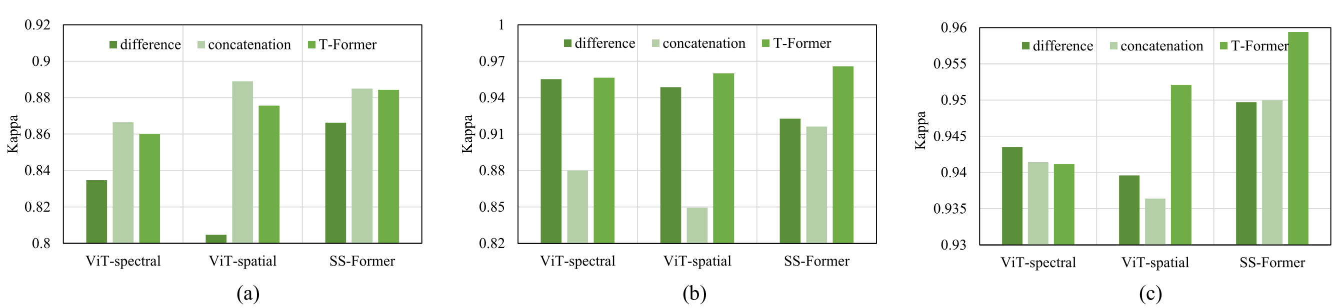


Fig. 11. Evaluation of the Kappa for T-Former on three datasets: (a) Farmland; (b) Santa Barbara; and (c) Bay Area.

1) Training Set Number: We selected seven training set numbers {100, 300, 500, 700, 900, 1100, 1300} to analyze their impact on SST-Former. Here, 100 presents that the number of changed training sets and the number of unchanged

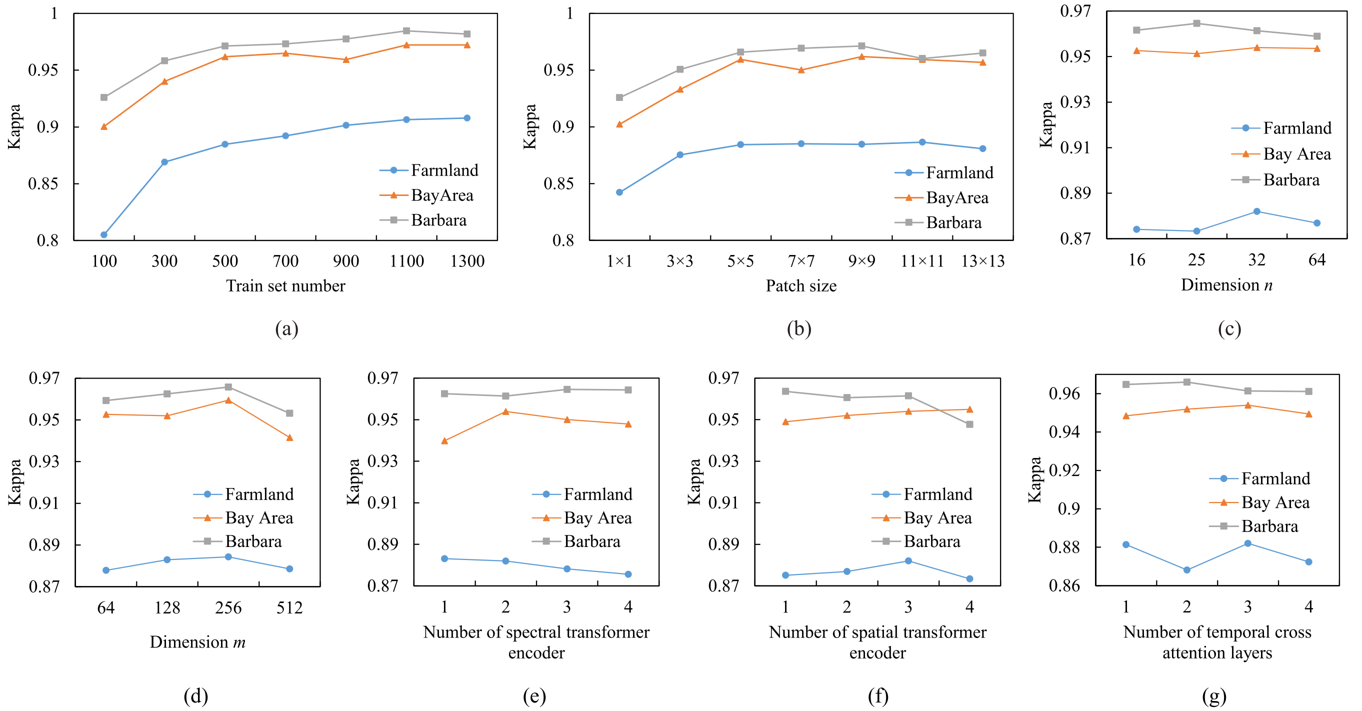


Fig. 12. Accuracy of SST-Former with different parameters on three datasets: (a) train set number; (b) patch size; (c) dimension n; (d) dimension m; (e) number of spectral transformer encoder; (f) number of spatial transformer encoder; and (g) number of temporal CA layers.

training sets are both 100. As shown in Fig. 12(a), the larger the number of training sets, the higher the CD accuracy is. As the number of training sets increases from 500 to 1300, the CD accuracy slowly improves.

2) Patch Size: We tested seven patch sizes $\{1 \times 1, 3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13\}$ to analyze their impact on SST-Former. In Fig. 12(b), we can see that with an increase in patch size, the CD accuracy first increases and then decreases. The patch sizes of 5×5, 7×7, 9×9, 11×11, and 13×13 have little difference on Kappa.

3) Dimension n: We experimented with $n = \{16, 25, 32, 64\}$ to find the optimal spectral transformer encoder dimension. The special dimension 25 is the pixel number of patch size 5×5. As shown in Fig. 12(c), the Kappa values using these four dimensions are similar, and 32 is almost the best dimension.

4) Dimension m: We experimented with $m = \{64, 128, 256, 512\}$, respectively, to find the optimal spatial transformer encoder dimension. As shown in Fig. 12(d), 256 is the best dimension.

5) Number of Spectral Transformer Encoders: To find the optimal number of spectral transformer encoders, four different numbers of encoders $\{1, 2, 3, 4\}$ are tested to find the optimal spectral transformer encoder number. As shown in Fig. 12(e), we find that different datasets have different optimal numbers of spectral transformer encoder layers. In this article, we selected 2 as the final number of spectral transformer encoders.

6) Number of Spatial Transformer Encoders: A spatial transformer is important to investigate the performance of the proposed method. Four different numbers of encoders $\{1, 2, 3, 4\}$ are tested to find the optimal spatial transformer encoder number. In Fig. 12(f), SST-Former with three encoders performs well on the three datasets.

7) Number of Temporal CA Layers: Similarly, the number of CAs in T-Former is also worth investigating. Four different numbers of CA layers $\{1, 2, 3, 4\}$ were tested to find the optimal temporal encoders. In Fig. 12(g), the best combined effect on the three datasets is achieved using three CA layers.

IV. Conclusion

HSIs contain considerable spectral sequence information, and multitemporal HSIs also contain a large amount of temporal sequence information. However, CNN and ViT focus on spatial features and ignore spectral and temporal sequence information. In this article, SST-Former is proposed for HSI-CD. SS-Former and T-Former are two parts of SST-Former. SS-Former is used to extract the spectral and spatial sequence features. T-Former can model the temporal sequence feature and find the change rule between different objects. We used small training datasets to validate that the SST-Former achieves the state-of-the-art performance on three datasets. SST-Former is not completely automatic and still requires manual labeling by experienced research for ground truth in a short time. Therefore, we will study strategy transformer structures with no labeled or fewer labeled datasets. This will be necessary for HSI-CD applications. At the same time, generating sets of big data for HSI-CD is a challenging and worthwhile task.

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REFERENCES

[1] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. Soc. Photo-Opt. Instrum. Eng.*, 1980, pp. 326–336.

[2] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, May 2012, doi: 10.1109/TGRS.2011.2171493.

[3] S. Liu, L. Bruzzone, F. Bovolo, M. Zanetti, and P. Du, "Sequential spectral change vector analysis for iteratively discovering and detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4363–4378, Aug. 2015, doi: 10.1109/TGRS.2015.2396686.

[4] M. Zanetti, F. Bovolo, and L. Bruzzone, "Rayleigh-Rice mixture parameter estimation via EM algorithm for change detection in multispectral images," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 5004–5016, Dec. 2015, doi: 10.1109/TIP.2015.2474710.

[5] C. Kwan, "Methods and challenges using multispectral and hyperspectral images for practical change detection applications," *Information*, vol. 10, no. 11, p. 353, Nov. 2019, doi: 10.3390/info10110353.

[6] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000, doi: 10.1109/36.843009.

[7] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 9, no. 1, pp. 62–66, Feb. 1979, doi: 10.1109/TSMC.1979.4310076.

[8] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008, doi: 10.1109/TGRS.2008.916643.

[9] Z. Chen and B. Wang, "Spectrally-spatially regularized low-rank and sparse decomposition: A novel method for change detection in multitemporal hyperspectral images," *Remote Sens.*, vol. 9, no. 10, p. 1044, Oct. 2017, doi: 10.3390/rs9101044.

[10] V. Ortiz-Rivera, M. Vélez-Reyes, and B. Roysam, "Change detection in hyperspectral imagery using temporal principal components," *Proc. SPIE*, vol. 6233, May 2006, Art. no. 623312, doi: 10.1117/12.667961.

[11] A. A. Nielsen and K. Conradsen, "Multivariate alteration detection (MAD) in multispectral, bi-temporal image data: A new approach to change detection studies," Dept. Math. Model., Tech. Univ. Denmark, Lyngby, Denmark, Tech. Rep. 1997-11, 1997, vol. 1, pp. 1–28. [Online]. Available: http://orbit.dtu.dk/en/publications/multivariate-alteration-detection-mad-in-multispectral-bitemporal-image-data-a-new-approach-to-change-detction-studies%288324cbcc-e13d-4770-b52b-c12ed66a4ba2%29.html

[12] A. A. Nielsen, "The regularized iteratively reweighted mad method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007, doi: 10.1109/TIP.2006.888195.

[13] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for hyperspectral change detection," in *Proc. 6th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2014, pp. 2858–2874, 2014, doi: 10.1109/WHISPERS.2014.8077588.

[14] L. Liu, D. Hong, L. Ni, and L. Gao, "Multilayer cascade screening strategy for semi-supervised change detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1926–1940, 2022, doi: 10.1109/JSTARS.2022.3150571.

[15] S. Liu, L. Bruzzone, F. Bovolo, and P. Du, "Unsupervised multitemporal spectral unmixing for detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2733–2748, May 2016, doi: 10.1109/TGRS.2015.2505183.

[16] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, 2006, doi: 10.1016/j.isprsjprs.2006.09.004.

[17] O. Ahlqvist, "Extending post-classification change detection using semantic similarity metrics to overcome class heterogeneity: A study of 1992 and 2001 U.S. National land cover database changes," *Remote Sens. Environ.*, vol. 112, no. 3, pp. 1226–1241, 2008, doi: 10.1016/j.rse.2007.08.012.

[18] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013, doi: 10.1016/j.isprsjprs.2013.03.006.

[19] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022, doi: 10.1109/TGRS.2021.3124913.

[20] D. Hong *et al.*, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2020, doi: 10.1109/TGRS.2020.3016820.

[21] X. Zheng, H. Sun, X. Lu, and W. Xie, "Rotation-invariant attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 4251–4265, 2022, doi: 10.1109/TIP.2022.3177322.

[22] X. Zheng, T. Gong, X. Li, and X. Lu, "Generalized scene classification from small-scale datasets with multitask learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, doi: 10.1109/TGRS.2021.3116147.

[23] Z. Wu *et al.*, "Scheduling-guided automatic processing of massive hyperspectral image classification on cloud computing architectures," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3588–3601, Jul. 2021, doi: 10.1109/TCYB.2020.3026673.

[24] A. Samat, E. Li, P. Du, S. Liu, Z. Miao, and W. Zhang, "CatBoost for RS image classification with pseudo label support from neighbor patches-based clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3038771.

[25] A. Samat, E. Li, W. Wang, S. Liu, C. Lin, and J. Abuduwaili, "Meta-XGBoost for hyperspectral image classification using extended MSER-guided morphological profiles," *Remote Sens.*, vol. 12, no. 12, p. 1973, Jun. 2020, doi: 10.3390/rs12121973.

[26] F. Luo, Z. Zou, J. Liu, and Z. Lin, "Dimensionality reduction and classification of hyperspectral image via multistructure unified discriminative embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2021.3128764.

[27] F. Luo, L. Zhang, B. Du, and L. Zhang, "Dimensionality reduction with enhanced hybrid-graph discriminant learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5336–5353, Aug. 2020, doi: 10.1109/TGRS.2020.2963848.

[28] Z. Wang, L. Zhuang, L. Gao, A. Marinoni, B. Zhang, and M. K. Ng, "Hyperspectral nonlinear unmixing by using plug-and-play prior for abundance maps," *Remote Sens.*, vol. 12, no. 24, p. 4117, Dec. 2020, doi: 10.3390/rs12244117.

[29] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5504017, doi: 10.1109/TGRS.2021.3065990.

[30] Z. Wu, W. Zhu, J. Chanussot, Y. Xu, and S. Osher, "Hyperspectral anomaly detection via global and local joint modeling of background," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3858–3869, Jul. 2019, doi: 10.1109/TSP.2019.2922157.

[31] Z. Wu, J. Sun, Y. Zhang, Z. Wei, and J. Chanussot, "Recent developments in parallel and distributed computing for remotely sensed big data processing," *Proc. IEEE*, vol. 109, no. 8, pp. 1282–1305, Aug. 2021, doi: 10.1109/JPROC.2021.3087029.

[32] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021, doi: 10.1109/TGRS.2020.3015157.

[33] Q. Li *et al.*, "Unsupervised hyperspectral image change detection via deep learning self-generated credible labels," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9012–9024, 2021, doi: 10.1109/JSTARS.2021.3108777.

[34] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018, doi: 10.1109/TGRS.2018.2849692.

[35] Y. Yang, J. Qu, S. Xiao, W. Dong, Y. Li, and Q. Du, "A deep multiscale pyramid network enhanced with spatial–spectral residual attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, doi: 10.1109/TGRS.2022.3161386.

[36] J. Lopez-Fandino, A. S. Garea, D. B. Heras, and F. Arguello, "Stacked autoencoders for multiclass change detection in hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 1906–1909, doi: 10.1109/IGARSS.2018.8518338.
[37] X. Zheng, X. Chen, X. Lu, and B. Sun, "Unsupervised change detection by cross-resolution difference learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2021.3079907.
[38] C. Zhao, H. Cheng, and S. Feng, "A spectral–spatial change detection method based on simplified 3-D convolutional autoencoder for multitemporal hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5507705, doi: 10.1109/LGRS.2021.3096526.
[39] A. Song, J. Choi, Y. Han, and Y. Kim, "Change detection in hyperspectral images using recurrent 3D fully convolutional networks," *Remote Sens.*, vol. 10, no. 11, p. 1827, Nov. 2018, doi: 10.3390/rs10111827.
[40] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, doi: 10.1109/TGRS.2021.3135567.
[41] J.-J. Wang, N. Dobigeon, M. Chabert, D.-C. Wang, J. Huang, and T.-Z. Huang, "CD-GAN: A robust fusion-based generative adversarial network for unsupervised change detection between heterogeneous images," 2022, *arXiv:2203.00948*.
[42] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510018, doi: 10.1109/TGRS.2021.3095899.
[43] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.
[44] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[45] C.-F. R. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2022, pp. 347–356, doi: 10.1109/ICCV48922.2021.00041.
[46] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229, doi: 10.1007/978-3-030-58452-8_13.
[47] Q. Wan, H. Ji, and L. Shen, "Self-attention based text knowledge mining for text detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5979–5988, doi: 10.1109/CVPR46437.2021.00592.
[48] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022, doi: 10.1109/TGRS.2022.3157671.
[49] W. Gedara Chaminda Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," 2022, *arXiv:2201.01293*.
[50] Y. Liu, J. Hu, X. Kang, J. Luo, and S. Fan, "Interactformer: Interactive transformer and CNN for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, doi: 10.1109/TGRS.2022.3183468.
[51] J. Chen and C. M. Ho, "MM-ViT: Multi-modal video transformer for compressed video action recognition," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 786–797, doi: 10.1109/WACV51458.2022.00086.
[52] Z. Zhang, X. Lu, G. Cao, Y. Yang, L. Jiao, and F. Liu, "ViT-YOLO: Transformer-based YOLO for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 2799–2808.
[53] A. Vaswani, "The transformer [historical overview]," *IEEE Ind. Appl. Mag.*, vol. 8, no. 1, pp. 8–15, Jan./Feb. 2002, doi: 10.1109/2943.974352.
[54] J. Gehring, M. Auli, D. Grangier, D. Yarats, and Y. N. Dauphin, "Convolutional sequence to sequence learning," 2017, *arXiv:1705.03122*.
[55] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
[56] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008, doi: 10.1080/01431160801950162.
[57] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019, doi: 10.1109/TIP.2018.2878958.

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